

name: Hero Model Reset
overview: "Validate your mental reset: separate three kinds of "model," use the locked hero as the fixed target, and pursue the simplest mechanism that can bend a selection curve—simulation is one test, not the only one. Most existing work stays; the change is sequencing and honesty about what each layer proves."
todos:
- id: memo-three-models
content: "Draft 1-page memo: phenomenological LPM vs L_net theory vs 538D generative — what each proves and does not"
status: completed
- id: hero-estimand-lock
content: Confirm hero spec in writing (16 quantile, winsor 0.01–0.99, min_minutes=20, poolq_loo) as fixed target for all model tests
status: completed
- id: lpm-layer-a
content: Run quadratic LPM on hero-filtered panel; record coef signs and overlay vs ventile bins
status: completed
- id: 538d-minimal-knockouts
content: "538D: $\lambda > 0$ vs $\lambda = 0$ and same 16-bin readout; try poolq_loo axis first, document axis if needed"
status: completed
- id: alex-side-by-side
content: Produce side-by-side figure + nesting-note limitation sentence for Alex (empirical hero + generative POC)
status: completed
- id: defer-stretch
content: Explicitly park assortativity sweeps and bin-for-bin LOO match as stretch, not v1 gate
status: completed
isProject: false

Hero inverted-U: model types, testing, and smart restart

Short answer: yes, your approach makes sense

You are doing the right thing by stepping back. The project already distinguishes **phenomenon** → **theory** → **generative POC** (`05_Model_Nesting_Note_v1.md` , `30_Alex_Gates_inform_status_outline.md`). Your confusion is mostly **which layer you are trying to “solve” at once**. Splitting those layers is the smart move.

1. What “mathematical model” means here (three different objects)

In everyday conversation we say “the model” as if there is one thing. In this project there are **three different objects**, and mixing them up is the main source of confusion. Each answers a different question. None of them, by itself, is the whole dissertation story.

The three layers in plain language

Layer	Plain-language name	What you are actually doing	What a successful result looks like
A. Phenomenological / reduced form	“Describe the curve in the data”	Fit a simple functional form to real player-season rows — e.g. does draft probability bend downward at the top when you plot it against leave-one-out pool quality?	You can show, on the locked hero spec , that draft rate rises and then dips (or is concave) when you smooth or bin the empirical panel. You get coefficients and a picture.
B. Structural / mechanism (theory)	“Explain why help and hurt can coexist”	Write down a story in symbols: net learning has a benefit part and a congestion part; selection may penalize crowding among viable peers, not just reward raw ability.	The story is internally consistent, maps to named objects in the code (<code>A_i</code> , congestion <code>L_C</code> , weight <code>λ</code>), and suggests falsifiable ingredient tests (e.g. “if congestion does not enter selection, the sim story fails”).
C. Generative / simulation (DGP)	“Build a fake league and see if the rule produces a bent curve”	Simulate rosters: draw ability, assign players to teams, rank by a selection score, draft the top <i>K</i> , then bin and plot who got selected vs a pool statistic.	Under a minimal rule set (score includes congestion, scarce slots), the shape of the simulated selection curve differs from talent-only selection — e.g. elite bins compress relative to ability-only.

Do not use one word (“the model”) for all three. Layer A is a **statistical summary** of MBB data. Layer B is **economic/statistical narrative** about mechanisms. Layer C is a **computerized thought experiment** that operationalizes part of Layer B.

Layer A — phenomenological (the hero and the quadratic LPM)

Question it answers: *Does an inverted-U-shaped (or concave) relationship between pool quality and draft probability show up in the real NCAA→NBA panel on a fixed, pre-specified definition of “quality”?*

What it is not: A claim that NBA teams literally maximize a particular formula, or that we have identified *why* the curve bends.

Concrete objects in the repo:

- **Hero plot** — binned mean draft rate (`Y_draft`) vs `poolq_loo`: each player’s leave-one-out mean teammate performance (teammates’ box-score quality with that player removed so they are not counted twice).
- **Quadratic LPM (linear probability model)** — regression on the same filtered rows:

$$Y_{\text{draft}} \sim \beta_0 + \beta_1 \cdot \text{poolq_loo} + \beta_2 \cdot \text{poolq_loo}^2$$

wired in `panel_build.py` (`draft_poolq_quadratic_coeffs` , ventile overlay in 530).

What “test” means here: Check sign and magnitude of `β2` (negative ⇒ concave / inverted-U tendency); compare the smooth curve to **ventile bin means** visually; optional out-of-sample checks. You are **fitting the phenomenon**, not simulating a league.

What Layer A proves: The stylized fact is **real on the locked estimand** (16 quantile bins, winsorized `poolq_loo` , `min_minutes=20` , etc.).

What it does not prove: Mechanism, causal effects, or that any NBA front office uses this score.

Layer B — structural / mechanism (theory spine)

Question it answers: *Why is it plausible that being on a “good” team pool could both help and hurt draft odds? — and **why selection is not the same as describing the environment.**

What it is: Prose and notation — not one estimated equation in v1. **Two parts, do not merge:**

Part 1 — Environment (`Lnet = B(·) - D(·)`): Peers confer benefit (B) and congestion (D) on net development / value in the environment. This **describes the pool**, not who gets drafted.

Part 2 — Selection (Alex score): `Si = Ai - λ·LC` is the **advancement rule** — who gets the scarce slot. Congestion enters **selection itself**, jointly with ability `Ai` . This is **not** the full reduced-form environment and **not** the hero regression.

Charles binding (Jul 2026): `3-Master_Plan/BINDING_Selection_is_its_own_step.md`

What “test” means here: Narrative consistency + sim **selection knockouts** — e.g. `λ = 0` (ability-only selection) and ask whether the generative story still holds.

What Layer B proves: A coherent **mechanism vocabulary** and falsifiable **directional** claims (congestion in selection can matter).

What it does not prove: Separate identification of `B(·)` and `D(·)` as functions of one axis in v1, or that the real NBA implements `Si` literally.

Layer C — generative simulation (the DGP)

Question it answers: *If we write down an explicit **data-generating process** — artificial players, teams, and a draft rule — does a **minimal** congestion term in the selection score change the **shape** of who gets picked when we bin by pool quality?*

DGP = data-generating process: the full recipe that **creates** synthetic rosters and draft outcomes, as opposed to fitting a curve to real outcomes after the fact.

Concrete stack in the repo (538D CELL 10):

1. Draw or assign **ability** for synthetic players.
2. **Soft assignment** to team types T_j (temperature τ) — who lands on which team context.
3. Compute a **selection score** (e.g. $S_i = A_i - \lambda \cdot L_C$).
4. **Top-K** draft — only K slots, so selection is scarce.
5. Plot **binned P(selected)** vs a pool feature (hero uses `poolq_loo`; sim may document if another axis is used for comparison — see Path II in the nesting note).

What “test” means here: Re-run the DGP with knobs toggled; compare **curves**, not regression R^2 on real data. The headline knockout: $\lambda > 0$ (congestion in score) vs $\lambda = 0$ (talent-only) — does the elite tail behave differently?

What Layer C proves: A **minimal artificial league** where talent-only selection is **not enough** and a congestion term in the score **can** bend the selection curve — proof-of-concept for Alex’s mechanism.

What it does not prove: Bin-for-bin replication of every hero ventile rate, or that the real draft **is** this sim.

How the three layers relate (read this before the diagram)

```
flowchart TB
    subgraph rung1 [Rung1_Empirical]
        Hero["Hero plot: mean draft rate vs poolq_loo ventiles"]
        LPM["Quadratic LPM overlay optional"]
    end
    subgraph rung2 [Rung2_Generative]
        Assign["Soft assignment: A_i, T_j, tau"]
        Score["Selection score: S_i = A_i - lambda L_C"]
        Select["Top-K draft slots"]
        SimPlot["Binned P selected vs pool feature"]
    end
    subgraph theory [Theory_spine]
        Lnet["L_net = B minus D"]
    end
    Hero --> LPM
    theory --> Score
    Assign --> Score --> Select --> SimPlot
    Hero -.->|"shape comparison honest axis"| SimPlot
    Lnet -.->|"frames why not only A"| Score
```

Reading the diagram:

- **Top path (Rung 1):** Start from **real MBB data** → hero bins + optional quadratic overlay. This is Layer **A**. No simulation required.
- **Middle path (Rung 2):** Start from **rules** → synthetic league → binned selection plot. This is Layer **C**. The sim **tests the rule**, not the regression.
- **Theory box:** Layer **B** sits **between** interpretation and design — it explains why selection might use $A - \lambda L_C$, not only A , but it does not by itself produce numbers until wired into Layer C (or decomposed empirically later at Rung 2.5).

Alex side-by-side (what v1 actually delivers): Layer **A** hero PNG + Layer **C** sim PNG, same **Y** (advancement / draft rate), **honest X** labels (empirical LOO quality vs sim bins on the best available matching axis), plus an explicit **limitation sentence** — not “derive the hero from a closed-form integral.”

One paragraph to keep in memory

Layer A can reproduce an inverted-U **without telling you why** — it is curve-fitting and binning on the real panel (**outcomes only**). **Layer B** splits **environment** (B – D) from **selection** (Alex score as its own step). **Layer C** runs the **selection step** in code and shows talent-only **selection** is insufficient. You need **A** for the empirical fact, **B** for the dissertation logic (environment ≠ selection), and **C** for the claim that **congestion in the selection rule** can bend curves — three jobs, three objects.

2. Does simulation test a formula?

Yes — when the formula is a generative rule, not when it is only a regression fit.

- **Simulation tests:** "If selection uses $S_i = A_i - \lambda \cdot L_C$ and rosters form this way, does **P(selected)** bend non-monotonically when we bin by a pool statistic?"
- **Regression tests:** "Does $P(\text{draft} \mid \text{poolq_loo})$ look quadratic in the **real panel**?" (already wired in 530)

Both are legitimate; they answer **different questions**. Simulation is **not** required to *draw* an inverted-U (a quadratic fit can do that). Simulation **is** required for the Wang-style claim: **talent-only selection is not enough; a congestion term in the score changes the curve** (already shown with $\lambda=0$ vs $\lambda>0$ in 538D).

3. Your two simplification strategies — which to use when

(a) Top-down: full mechanism → drop until it breaks

Best for: Layer B/C — Alex score, assignment, roster formation.

Procedure: 1. Start with the **full generative stack** you already built (ability draw, soft match to T_j , congestion in score, top-K). 2. Fix **one empirical anchor** (hero spec below). 3. Remove or zero one ingredient at a time and record **what breaks**: - $\lambda = 0$ → talent-only (should flatten / fail to show congestion story) - no soft assignment (hard assortative chop) → changes pool overlap / tail bins - no L_C term, only A_i → selection on ability alone

This is **mechanism identification in the sim**, not coefficient estimation on real draft data.

(b) Bottom-up: minimal curve → decompose

Best for: Layer A — describing the hero.

Procedure: 1. Start with **simplest curve**: $E[Y \mid Q] = \beta_0 + \beta_1 Q + \beta_2 Q^2$ with $\beta_2 < 0$ (530 LPM). 2. Decompose **Q** (poolq_loo) into interpretable pieces only when moving to Rung 2.5: - $\text{poolq_loo} \approx$ quality column (B-ish mix) - $\text{crowding_smooth} \approx$ congestion column (D-ish) — CELL 5d in `530_sports_pipeline.ipynb`

Recommendation: Use **both**, in order: **bottom-up for the hero (A)**, **top-down for the sim (C)**. They are not competing; they are different rungs.

4. Assortative grouping, τ , etc. — simulation details or essentials?

Mostly simulation / DGP details — not the core mathematical claim.

Knob	Role	Needed for v1 Alex side-by-side?
λ / congestion in score	Core mechanism ingredient	Yes
Top-K selection	Scarcity of draft slots	Yes (conceptually)
Soft assignment (τ , T_j)	How players land on teams	Helpful for realistic pool overlap; not the headline theorem
Assortative sort-and-chop benchmark	537 overlay	Diagnostic only (Plot A overlap)
Empirical team distribution on/off	Faithful 538 sweeps	Stretch / appendix — not gate for “minimal POC”

Path II (locked in nesting note): generative figure may use **pool mean** on X while empirical hero uses `poolq_loo`. Say that aloud; do not over-fit assortativity until the **minimal** story works.

Stretch goal (parallel, not v1 gate): match hero on `poolq_loo` with same binning (`alex_side_by_side_v0/` spec: 16 quantile, winsor 0.01–0.99, `min_minutes=20`).

5. Do you need simulation? Why?

Goal	Need sim?
Show inverted-U exists in MBB data	No — hero is done
Fit a smooth curve through bins	No — quadratic LPM
Explain <i>why</i> help and hurt can coexist	Partly prose ($L_{net} = B - D$)
Show congestion in selection is a plausible minimal ingredient	Yes — 538D
Prove identification of B(Q) and D(Q) separately on one axis	No — explicitly deferred in v1

Why sim exists in your project: Alex’s score is a **rule for an artificial league**, not a claim that real NBA teams optimize S_i . Simulation is how you **operationalize and stress-test** that rule. It is the supplement; the hero is the spine.

6. What is probably a “keeper” vs what can wait

Keep (solid foundation): - Locked hero triplet + 530 panel/LOO pipeline - Nesting note + Alex brief axis honesty - 538D CELL 10 generative lab ($S = A - \lambda L_C$, talent-only control) - Sensitivity lessons in `Pertinent_Thoughts_Scout.md` ($min_minutes$, quantile vs equal-width)

Defer / shrink (avoid re-blocking): - Bin-for-bin LOO replication as **v1 requirement** - Large faithful-538 sweeps on Rivanna until minimal sim curve is locked - HS binning, multiplicative rewrite, full $B(Q)-D(Q)$ estimation - Extra knobs (assortativity grids, empirical team cloning) until minimal λ + K story is documented in one page

7. Recommended sequence (slow, smart)

1. **Write one page “three models” memo** (you + agent): hero = Rung 1; LPM = descriptive; 538D = Rung 2; explicit non-claims.
2. **Freeze hero estimand** (already): 16 quantile, ppm z-scored, winsor (0.01, 0.99), $min_minutes=20$, 2011–2021, $poolq_loo$, draftee filter off.
3. **Layer A check (1 hour):** Run quadratic LPM on same filtered rows; confirm negative quadratic term aligns with top ventile dip.
4. **Layer C minimal (538D):** One knob sheet: $\lambda>0$, $\lambda=0$, same bin count as hero; plot on **best available X** (try $poolq_loo$ first; document if only pool-mean bends).
5. **Top-down knockouts:** Zero λ , simplify assignment; record which kills non-monotone shape.
6. **Alex slide:** side-by-side PNGs + one limitation sentence from nesting note §4.
7. **Only then** revisit assortativity / faithful sweeps if LOO match still matters scientifically.

8. One-sentence answers to your numbered questions

1. **Recreate inverted-U:** Yes — via **quadratic reduced form** (easy) and/or **generative selection with congestion** (mechanism); full **B-D** decomposition is theory, not yet a single estimated formula.
2. **What is a model / how tested:** Three layers (table above); simulation tests **generative rules**, regression tests **empirical curve**, theory frames **interpretation**.
3. **Top-down vs bottom-up:** Use **both** — bottom-up for hero fit, top-down for sim mechanism stripping.
4. **Assortativity etc.:** Mostly **simulation plumbing**; not the v1 theorem. Sim is needed for **minimal mechanism POC**, not for proving the hero exists.
5. **Reset smart? Yes.** Keep infrastructure; re-order claims around rungs; stop requiring one formula to do all jobs.
6. **Approach correct? Yes**, with one guardrail: **do not make bin-for-bin LOO sim match the gate for “we have a model.”** That is north-star R&D, not the definition of success for the next Alex conversation.